COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010-CPE21S4	**SECTION:**	CPE21S4
DATE PERFORMED:	3/09/2026	**DATE SUBMITTED:**	22/08/2026
NAME:	Jullian Edrain R. Sevilla	**INSTRUCTOR:**	Engr. JIMLORD QUEJADO

ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

What procedures did you perform?
Answer the following questions:

1. What is a sorting algorithm?
   - A sorting algorithm is just a step by step way to put things in order, like arranging numbers from smallest to biggest, or names from A to Z

2. Where can sorting algorithms be used?
   - Computer files and Contacts

3. Explain the different types of sorting algorithms.
   - Bubble sort: It checks numbers next to each other and swaps them if they are in the wrong order until the whole list is sorted.
   - Selection sort: It finds the smallest number first, puts it in front, then finds the next smallest and puts it next.
   - Insertion sort: It takes each number one by one and slides it into the right spot, like how you sort cards in your hand.

4. Give sample programs in C++ that use sorting algorithms, specifically selection sort, insertion sort, and bubble sort. Explain how the programs work.
   - Bubble sort: Contacts sorts names A to Z.
   - Selection sort: Item price on online shop, from cheapest to most expensive.
   - Insertion sort: Gmail inbox, straight to the right place and by date.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

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Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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